



Product Specification

M240UAN02.1

AU OPTRONICS CORPORATION

() Preliminary Specification
(V) Final Specification

Module	24" Color TFT-LCD
Model Name	M240UAN02.1

<table><tr><td>Customer</td><td>Date</td></tr><tr><td>_____</td><td>_____</td></tr><tr><td>Approved by</td><td></td></tr><tr><td>_____</td><td>_____</td></tr></table>	Customer	Date	_____	_____	Approved by		_____	_____	<table><tr><td>Approved by</td><td>Date</td></tr><tr><td><u>Howard Lee</u></td><td><u>Dec 29, 2017</u></td></tr><tr><td>Prepared by</td><td>Date</td></tr><tr><td><u>Louis Jung</u></td><td><u>Dec 29, 2017</u></td></tr></table>	Approved by	Date	<u>Howard Lee</u>	<u>Dec 29, 2017</u>	Prepared by	Date	<u>Louis Jung</u>	<u>Dec 29, 2017</u>
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Note: This Specification is subject to change without notice.	AU Optronics corporation																

Contents

Revised	3
1 Handling Precautions	5
2 General Description	6
2.1 Display Characteristics	6
2.2 Absolute Maximum Rating of Environment	7
2.3 Optical Characteristics	8
2.4 Mechanical Characteristics	13
3 TFT-LCD Module	14
3.1 Block Diagram	14
3.2 Interface Connection	15
3.2.1 Connector Type	15
3.2.2 Connector Pin Assignment	15
3.3 Electrical Characteristics	18
3.3.1 Absolute Maximum Rating	18
3.3.2 Recommended Operating Condition	18
3.4 Signal Characteristics	19
3.4.1 LCD Pixel Format	19
3.4.2 eDP Data Format	19
3.4.3 Color versus Input Data	20
3.4.4 eDP Specification (Follow as VESA DisplayPort Standard Version 1.1)	21
3.4.5 Input Timing Specification	24
3.4.6 Input Timing Diagram	25
3.5 Power ON/OFF Sequence	26
4 Backlight Unit	28
4.1 Block Diagram	28
4.2 Interface Connection	29
4.2.1 Connector Type	29
4.2.2 Connector Pin Assignment	31
4.3 Electrical Characteristics	32
4.3.1 Absolute Maximum Rating	32
4.3.2 Recommended Operating Condition	32
5 Reliability Test	34
6 Shipping Label	36
7 Mechanical Characteristics	37
8 Packing Specification	39
8.1 Packing Flow	39
8.2 Pallet and shipment information	40
9 Design Guide for System	41

Record of Revision

Version	Date	Page	Old description	New Description	Remark																																																																																																																																						
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Product Specification

M240UAN02.1

AU OPTRONICS CORPORATION

		24	3.4.6 Input Timing Diagram The diagram illustrates the input timing characteristics of the 7805 voltage regulator. It shows the relationship between the input voltage (V_{in}) and the output voltage (V_{out}) under various conditions. Key parameters shown include: Delay Time (t_d): The time interval from the start of the input voltage step to the beginning of the output voltage response. Regulation Time (t_r): The time interval from the end of the input voltage step to the point where the output voltage has settled to its final value within a specified tolerance. Ripple Rejection: The ability of the regulator to reject AC noise or ripple present on the input line. <table><tr><th>Load Current Range</th><th>Input Voltage Range</th><th>Output Voltage Range</th><th>Parameter</th></tr><tr><td>0 - 100 mA</td><td>10 - 25 V</td><td>5.0 - 5.2 V</td><td>Delay Time (t_d)</td></tr><tr><td>100 - 200 mA</td><td>10 - 25 V</td><td>5.0 - 5.2 V</td><td>Delay Time (t_d)</td></tr><tr><td>200 - 300 mA</td><td>10 - 25 V</td><td>5.0 - 5.2 V</td><td>Delay Time (t_d)</td></tr><tr><td>300 - 400 mA</td><td>10 - 25 V</td><td>5.0 - 5.2 V</td><td>Delay Time (t_d)</td></tr><tr><td>400 - 500 mA</td><td>10 - 25 V</td><td>5.0 - 5.2 V</td><td>Delay Time (t_d)</td></tr><tr><td>500 - 600 mA</td><td>10 - 25 V</td><td>5.0 - 5.2 V</td><td>Delay Time (t_d)</td></tr><tr><td>600 - 700 mA</td><td>10 - 25 V</td><td>5.0 - 5.2 V</td><td>Delay Time (t_d)</td></tr><tr><td>700 - 800 mA</td><td>10 - 25 V</td><td>5.0 - 5.2 V</td><td>Delay Time (t_d)</td></tr><tr><td>800 - 900 mA</td><td>10 - 25 V</td><td>5.0 - 5.2 V</td><td>Delay Time (t_d)</td></tr><tr><td>900 - 1000 mA</td><td>10 - 25 V</td><td>5.0 - 5.2 V</td><td>Delay Time (t_d)</td></tr></table> <table><tr><th>Load Current Range</th><th>Input Voltage Range</th><th>Output Voltage Range</th><th>Parameter</th></tr><tr><td>0 - 100 mA</td><td>10 - 25 V</td><td>5.0 - 5.2 V</td><td>Regulation Time (t_r)</td></tr><tr><td>100 - 200 mA</td><td>10 - 25 V</td><td>5.0 - 5.2 V</td><td>Regulation Time (t_r)</td></tr><tr><td>200 - 300 mA</td><td>10 - 25 V</td><td>5.0 - 5.2 V</td><td>Regulation Time (t_r)</td></tr><tr><td>300 - 400 mA</td><td>10 - 25 V</td><td>5.0 - 5.2 V</td><td>Regulation Time (t_r)</td></tr><tr><td>400 - 500 mA</td><td>10 - 25 V</td><td>5.0 - 5.2 V</td><td>Regulation Time (t_r)</td></tr><tr><td>500 - 600 mA</td><td>10 - 25 V</td><td>5.0 - 5.2 V</td><td>Regulation Time (t_r)</td></tr><tr><td>600 - 700 mA</td><td>10 - 25 V</td><td>5.0 - 5.2 V</td><td>Regulation Time (t_r)</td></tr><tr><td>700 - 800 mA</td><td>10 - 25 V</td><td>5.0 - 5.2 V</td><td>Regulation Time (t_r)</td></tr><tr><td>800 - 900 mA</td><td>10 - 25 V</td><td>5.0 - 5.2 V</td><td>Regulation Time (t_r)</td></tr><tr><td>900 - 1000 mA</td><td>10 - 25 V</td><td>5.0 - 5.2 V</td><td>Regulation Time (t_r)</td></tr></table>	Load Current Range	Input Voltage Range	Output Voltage Range	Parameter	0 - 100 mA	10 - 25 V	5.0 - 5.2 V	Delay Time (t_d)	100 - 200 mA	10 - 25 V	5.0 - 5.2 V	Delay Time (t_d)	200 - 300 mA	10 - 25 V	5.0 - 5.2 V	Delay Time (t_d)	300 - 400 mA	10 - 25 V	5.0 - 5.2 V	Delay Time (t_d)	400 - 500 mA	10 - 25 V	5.0 - 5.2 V	Delay Time (t_d)	500 - 600 mA	10 - 25 V	5.0 - 5.2 V	Delay Time (t_d)	600 - 700 mA	10 - 25 V	5.0 - 5.2 V	Delay Time (t_d)	700 - 800 mA	10 - 25 V	5.0 - 5.2 V	Delay Time (t_d)	800 - 900 mA	10 - 25 V	5.0 - 5.2 V	Delay Time (t_d)	900 - 1000 mA	10 - 25 V	5.0 - 5.2 V	Delay Time (t_d)	Load Current Range	Input Voltage Range	Output Voltage Range	Parameter	0 - 100 mA	10 - 25 V	5.0 - 5.2 V	Regulation Time (t_r)	100 - 200 mA	10 - 25 V	5.0 - 5.2 V	Regulation Time (t_r)	200 - 300 mA	10 - 25 V	5.0 - 5.2 V	Regulation Time (t_r)	300 - 400 mA	10 - 25 V	5.0 - 5.2 V	Regulation Time (t_r)	400 - 500 mA	10 - 25 V	5.0 - 5.2 V	Regulation Time (t_r)	500 - 600 mA	10 - 25 V	5.0 - 5.2 V	Regulation Time (t_r)	600 - 700 mA	10 - 25 V	5.0 - 5.2 V	Regulation Time (t_r)	700 - 800 mA	10 - 25 V	5.0 - 5.2 V	Regulation Time (t_r)	800 - 900 mA	10 - 25 V	5.0 - 5.2 V	Regulation Time (t_r)	900 - 1000 mA	10 - 25 V	5.0 - 5.2 V	Regulation Time (t_r)	
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1 Handling Precautions

- 1) Since polarizer is easily damaged, do not touch or press the surface of polarizer with hand.
- 2) Be sure to turn off power supply when inserting or disconnecting from input connector.
- 3) Wipe off water drop immediately. Long contact with water may cause discoloration or spots.
- 4) When the panel surface is soiled, wipe it with absorbent cotton or other soft cloth.
- 5) Since the panel is made of glass, it may break or crack if dropped or bumped on hard surface.
- 6) Since CMOS LSI is used in this module, take care of static electricity and insure human earth when handling.
- 7) Do not open or modify the Module Assembly.
- 8) Do not press the reflector sheet at the back of the module to any directions.
- 9) In case a TFT-LCD Module has to be put back into the packing container slot after once it was taken out from the container, do not press the center of the LED lightbar edge. Otherwise the TFT-LCD Module may be damaged.
- 10) Insert or pull out the interface connector, be sure not to rotate nor tilt it of the TFT-LCD Module.
- 11) Do not twist nor bend the TFT -LCD Module even momentary. It should be taken into consideration that no bending/twisting forces are applied to the TFT-LCD Module from outside. Otherwise the TFT-LCD Module may be damaged.
- 12) Please avoid touching COF position while you are doing mechanical design.
- 13) When storing modules as spares for a long time, the following precaution is necessary: Store them in a dark place. Do not expose the module to sunlight or fluorescent light. Keep the temperature between 5°C and 35°C at normal humidity.
- 14) Do not apply the same pattern for a long time, it will enhance relevant defect.

2 General Description

This specification applies to the 24 inch wide Color a-Si TFT-LCD Module M240UAN02.1. The display supports the WUXGA - 1920(H) x 1200(V) screen format and 1.07BM colors (RGB 8 bits+Hi-FRC). The input interface is 4-lanes eDP and this module doesn't contain an driver board for backlight.

2.1 Display Characteristics

The following items are characteristics summary on the table under 25°C condition:

ITEMS	Unit	SPECIFICATIONS
Screen Diagonal	[mm]	611.32 (24.07")
Active Area	[mm]	518.4 (H) x 324.0 (V)
Pixels H x V	-	1920(x3) x 1200
Pixel Pitch	[um]	270 (per one triad) x270
Pixel Arrangement	-	R.G.B. Vertical Stripe
Display Mode	-	AHVA Mode (Advanced Hyper-Viewing Angle), Normally Black
White Luminance (Center)	[cd/m ²]	350 (Typ.)
Contrast Ratio	-	1000 (Typ.)
Color Gamut	-	Adobe 99%
Response Time	[msec]	14 (Typ., GTG)
Power Consumption (LCD Module + Backligh unit)	[Watt]	21.6W(Typ.) LCD module : PDD (Typ.)=5.4W @ White pattern,Fv=60 Hz Backlight unit : P _{BLU} (Typ.) =16.2W @Is=90mA
Weight	[Grams]	2420 +/- 50
Outline Dimension	[mm]	528.2(H) x 342.4(V) x 13.6(D) Typ.
Electrical Interface	-	4-lanes eDP , 10bits RGB data input
Support Color	-	1.07B colors (RGB 8 bits+Hi-FRC)
Surface Treatment	-	Anti-Glare, SAG25
Temperature Range Operating Storage (Shipping)	[°C] [°C]	0 to +50 -20 to +60
RoHS Compliance	-	RoHS Compliance
TCO Compliance	-	TCO 7.0 Compliance

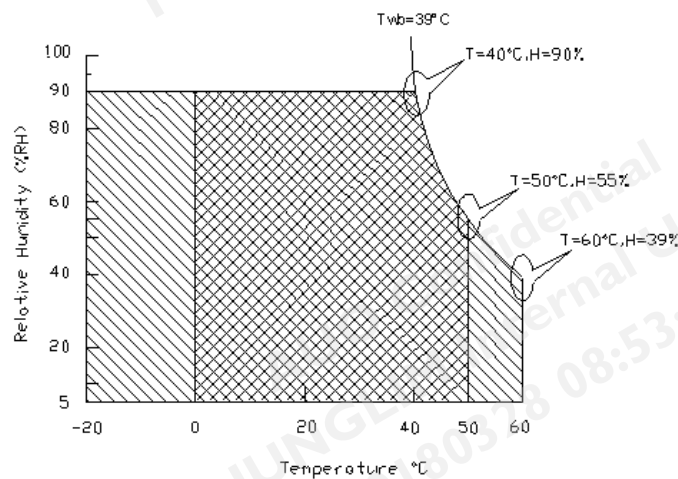
2.2 Absolute Maximum Rating of Environment

Permanent damage may occur if exceeding the following maximum rating.

Symbol	Description	Min.	Max.	Unit	Remark
TOP	Operating Temperature	0	+50	[°C]	Note 2-1
TGS	Glass surface temperature (operation)	0	+65	[°C]	Note 2-1 Function judged only
HOP	Operation Humidity	5	90	[%RH]	Note 2-1
TST	Storage Temperature	-20	+60	[°C]	
HST	Storage Humidity	5	90	[%RH]	

Note 2-1: Temperature and relative humidity range are shown as the below figure.

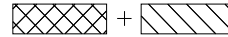
1. 90% RH Max ($T_a \leq 39^\circ\text{C}$)
2. Max wet-bulb temperature at 39°C or less. ($T_a \leq 39^\circ\text{C}$)
3. No condensation



Operating Range



Storage Range



2.3 Optical Characteristics

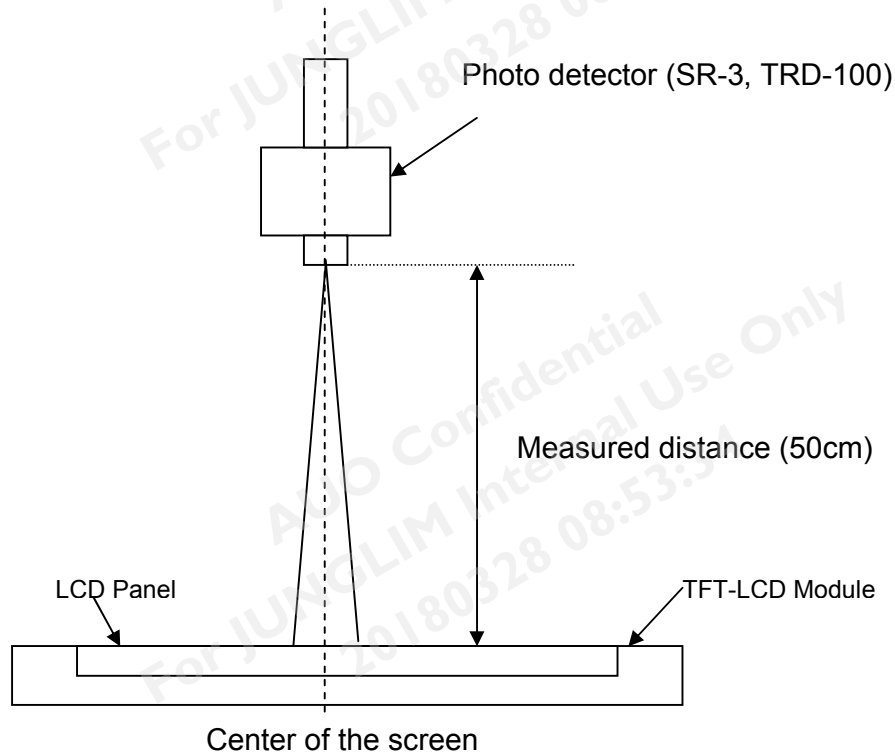
The optical characteristics are measured on the following test condition.

Test Condition:

1. Equipment setup: Please refer to **Note 2-2**.
2. Panel Lighting time: 30 minutes
3. VDD=12V, Fv=60Hz, Is=90mA, Ta=25°C

Symbol	Description		Min.	Typ.	Max.	Unit	Remark
L _w	White Luminance (Center of screen)		280	350	-	[cd/m2]	Note 2-2 By SR-3
L _{uni}	Luminance Uniformity (9 points)		75	80	-	[%]	Note 2-3 By SR-3
CR	Contrast Ratio (Center of screen)		600	1000	-	-	Note 2-4 By SR-3
θ _R	Horizontal Viewing Angle (CR=10)	Right	75	89	-	[degree]	Note 2-5 By SR-3
θ _L		Left	75	89	-		
Φ _H	Vertical Viewing Angle (CR=10)	Up	70	89	-		
Φ _L		Down	70	89	-		
θ _R	Horizontal Viewing Angle (CR=5)	Right	75	89	-		
θ _L		Left	75	89	-		
Φ _H	Vertical Viewing Angle (CR=5)	Up	70	89	-		
Φ _L		Down	70	89	-		
T _{GTG}	Response Time	Gray To Gray	-	14	-	[msec]	Note 2-6 By TRD-100
R _x	Color Coordinates (CIE 1931)	Red x	0.656	0.686	0.716	-	By SR-3
R _y		Red y	0.280	0.310	0.340		
G _x		Green x	0.179	0.209	0.239		
G _y		Green y	0.682	0.712	0.742		
B _x		Blue x	0.119	0.149	0.179		
B _y		Blue y	0.020	0.050	0.080		
W _x		White x	0.283	0.313	0.343		
W _y		White y	0.299	0.329	0.359		
Adobe RGB coverage ratio				99		[%]	By SR-3
CT	Crosstalk		-	-	1.5	[%]	Note 2-7 By SR-3
F _{dB}	Flicker (Center of screen)		-	-	-20	[dB]	Note 2-8 By SR-3

Note 2-2: Equipment setup :

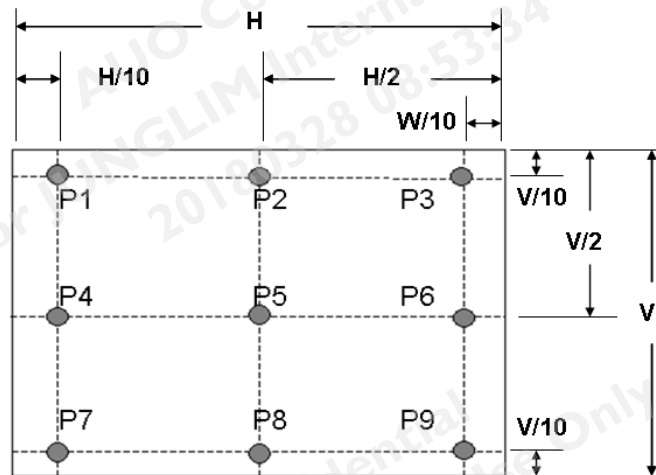


Note 2-3: Luminance Uniformity Measurement

Definition:

$$\text{Luminance Uniformity} = \frac{\text{Minimum Luminance of 9 Points (P1 ~ P9)}}{\text{Maximum Luminance of 9 Points (P1 ~ P9)}}$$

a. Test pattern: White Pattern



Note 2-4: Contrast Ratio Measurement

Definition:

$$\text{Contrast Ratio} = \frac{\text{Luminance of White pattern}}{\text{Luminance of Black pattern}}$$

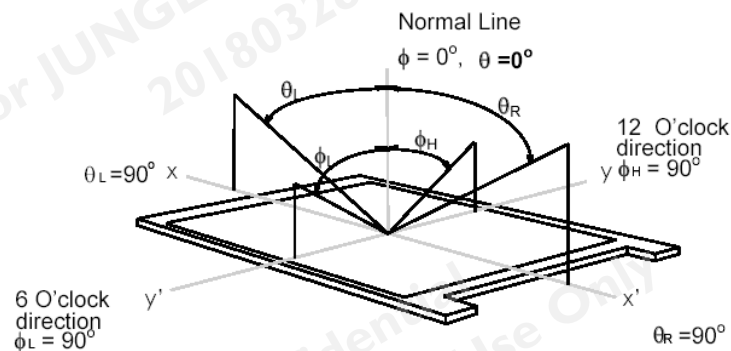
a. Measured position: Center of screen (P5) & perpendicular to the screen
($\theta = \phi = 0^\circ$)

Note 2-5: Viewing angle measurement

Definition: The angle at which the contrast ratio is greater than 10 & 5 .

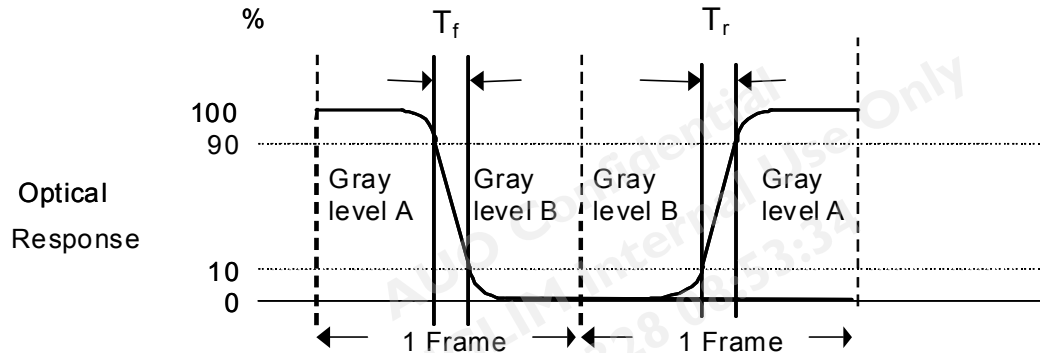
a. Horizontal view angle: Divide to left & right (θ_L & θ_R)

Vertical view angle: Divide to up & down (ϕ_H & ϕ_L)



Note 2-6: Response time measurement

The output signals of photo detector are measured when the input signals are changed from “Gray level A” to “Gray level B” (falling time, T_f), and from “Gray level B” to “Gray level A” (rising time, T_r), respectively. The response time is interval between the 10% and 90% of optical response.



The gray to gray response time is defined as the following table.

Gray Level to Gray Level		Target gray level				
		L0	L255	L511	L767	L1023
Start gray level	L0					
	L255					
	L511					
	L767					
	L1023					

■ T_{GTG_typ} is the total average time at rising time and falling time of gray to gray.

Note 2-7: Crosstalk measurement

Definition:

$$CT = \text{Max. } (CT_H, CT_V);$$

Where

a. Maximum Horizontal Crosstalk :

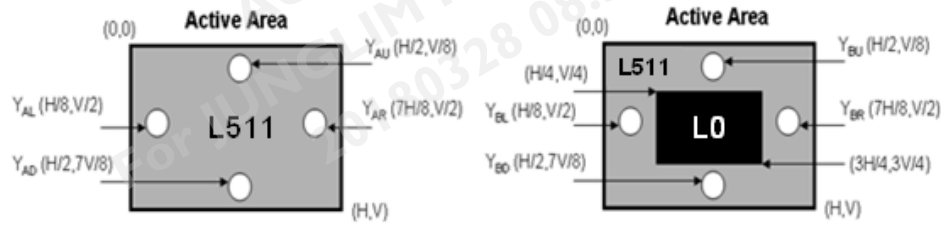
$$CT_H = \text{Max. } (|Y_{BL} - Y_{AL}| / Y_{AL} \times 100\%, |Y_{BR} - Y_{AR}| / Y_{AR} \times 100\%);$$

Maximum Vertical Crosstalk:

$$CT_V = \text{Max. } (|Y_{BU} - Y_{AU}| / Y_{AU} \times 100\%, |Y_{BD} - Y_{AD}| / Y_{AD} \times 100\%);$$

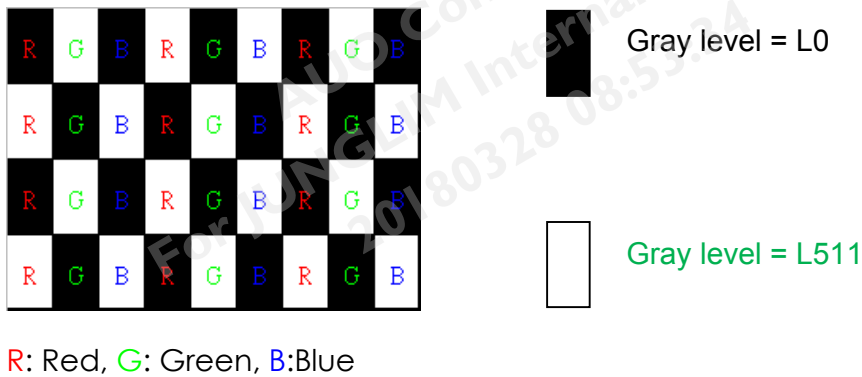
b. Y_{AU} , Y_{AD} , Y_{AL} , Y_{AR} = Luminance of measured location without Black pattern

Y_{BU} , Y_{BD} , Y_{BL} , Y_{BR} = Luminance of measured location with Black pattern



Note 2-8: Flicker measurement

a. Test pattern: It is listed as following.



b. Measured position: Center of screen (P5) & perpendicular to the screen
($\theta = \phi = 0^\circ$)

2.4 Mechanical Characteristics

Symbol	Description	Min.	Max.	Unit	Remark
P _{bc}	Backside Compression	2.5	-	[Kgf]	Note 2-9

Note 2-9: 1. Test condition:

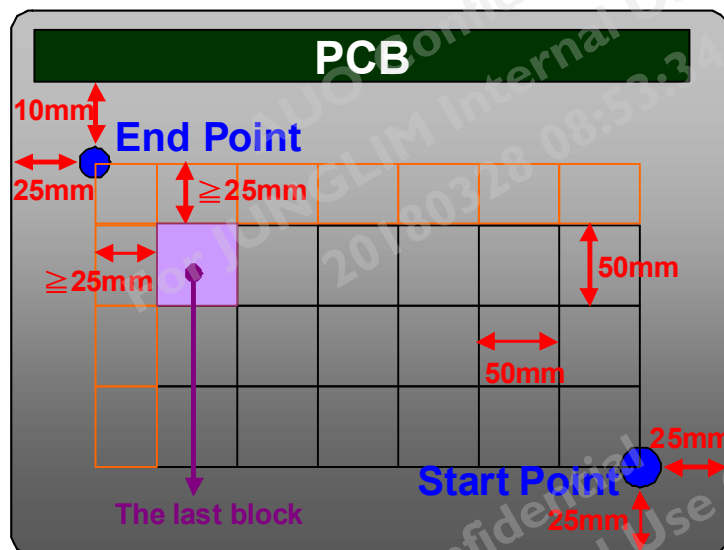
- (1) Ambient illumination : 10 ~ 15 lux
- (2) Test pattern : Gray (L511)
- (3) Distance of judgment : 30 cm from surface of LCD panel
- (4) Viewing angle of judgment : Front
- (5) Test jig : Push-Pull Gauge & Flat Head (Φ2)

2. Test Method:

The point is at a distance from right-downside 25mm x 25mm defined as the Start Point of Measure Points, and the point is at a distance 25mm from left-side & around 10mm from PCB defined as the End Point.

Align 50mm x 50mm block from Start Point on the Bezel Back, and the corners of each block are Measure Points.

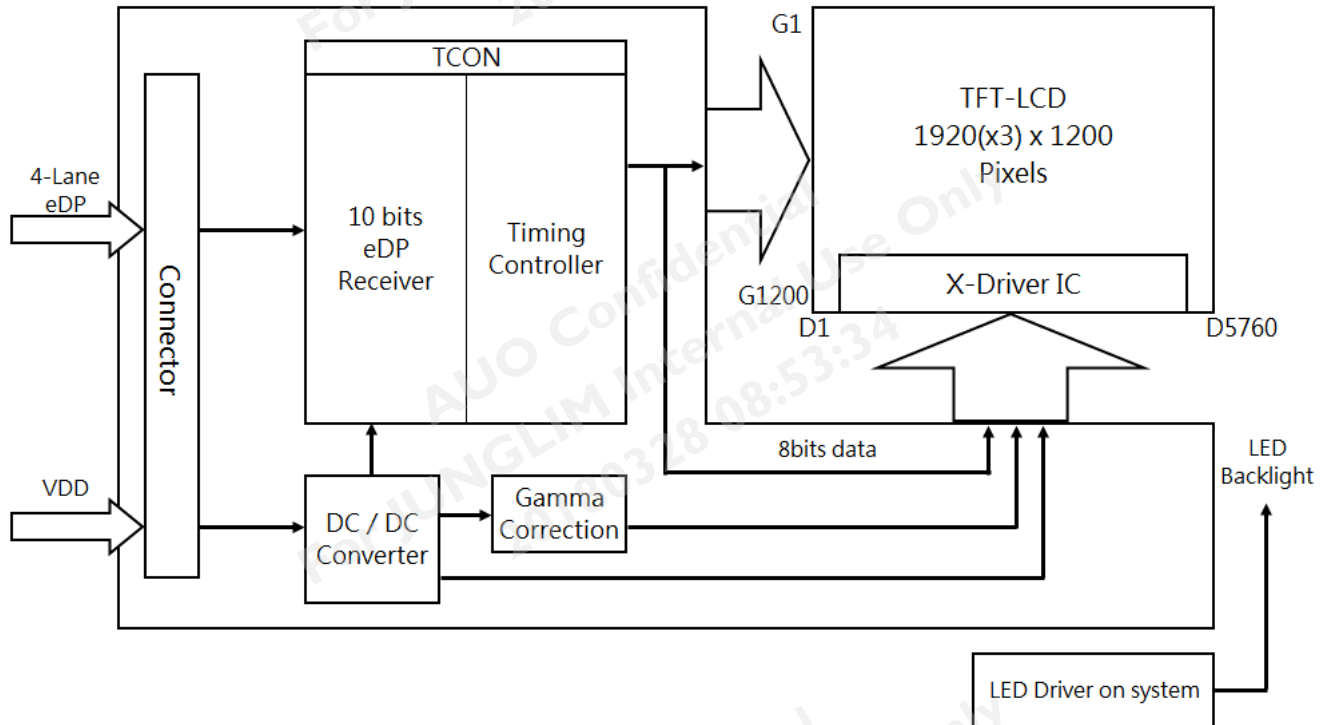
If the distance from the last block to each side of the End Point ≥ 25 mm, add other blocks to make sure that most area of Bezel Back can be measured.



3 TFT-LCD Module

3.1 Block Diagram

The following shows the block diagram of the 24 inch Color TFT-LCD Module.



3.2 Interface Connection

3.2.1 Connector Type

TFT-LCD Connector	Manufacturer	JAE	P-TWO	STARCONN (CHIEF LAND)
	Part Number	FI-RTE51SZ-HF	187059-5122	115E51-0000RA-M3-R
Mating Connector	Manufacturer	JAE or Compatible		
	Part Number	FI-RE51CL (Locked Type)		

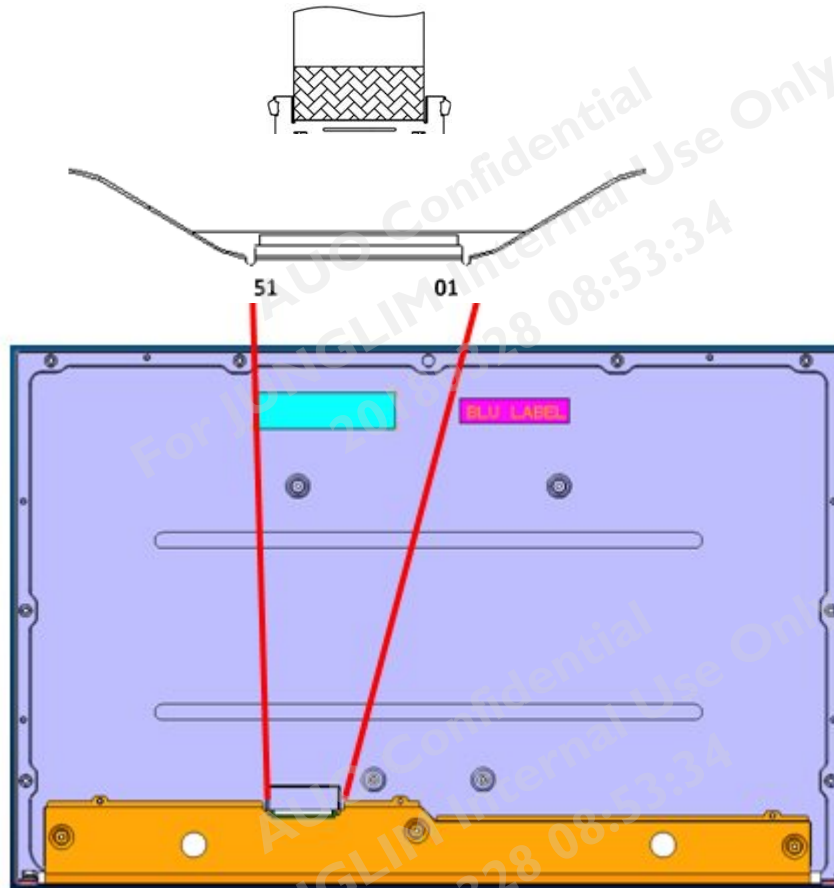
3.2.2 Connector Pin Assignment

PIN #	Symbol	Description	Remark
1	VDD	Power +12V	
2	VDD	Power +12V	
3	VDD	Power +12V	
4	VDD	Power +12V	
5	VDD	Power +12V	
6	GND	Ground	
7	GND	Ground	
8	GND	Ground	
9	NC	No connection (for AUO test only. Do not connect)	
10	NC	No connection (for AUO test only. Do not connect)	
11	NC	No connection (for AUO test only. Do not connect)	
12	NC	No connection (for AUO test only. Do not connect)	
13	NC	No connection (for AUO test only. Do not connect)	
14	NC	No connection (for AUO test only. Do not connect)	
15	NC	No connection (for AUO test only. Do not connect)	
16	NC	No connection (for AUO test only. Do not connect)	
17	GND	Ground	
18	Lane3_N	Negative eDP differential data input	
19	Lane3_P	Positive eDP differential data input	



20	GND	Ground	
21	Lane2_N	Negative eDP differential data input	
22	Lane2_P	Positive eDP differential data input	
23	GND	Ground	
24	Lane1_N	Negative eDP differential data input	
25	Lane1_P	Positive eDP differential data input	
26	GND	Ground	
27	Lane0_N	Negative eDP differential data input	
28	Lane0_P	Positive eDP differential data input	
29	GND	Ground	
30	AUX_CH_P	Positive AUX Channel differential data input	
31	AUX_CH_N	Negative AUX Channel differential data input	
32	GND	Ground	
33	GND	DG By Pass For BenQ Only	
34	GND	Ground	
35	NC	Do not connect	
36	NC	Do not connect	
37	GND	Ground	
38	NC	Do not connect	
39	NC	Do not connect	
40	GND	Ground	
41	NC	Do not connect	
42	NC	Do not connect	
43	GND	Ground	
44	NC	Do not connect	
45	NC	Do not connect	
46	GND	Ground	
47	NC	Do not connect	
48	NC	Do not connect	

49	GND	Ground	
50	HPD	Hot plug detection	
51	NC	Do not connect	



3.3 Electrical Characteristics

3.3.1 Absolute Maximum Rating

Permanent damage may occur if exceeding the following maximum rating.

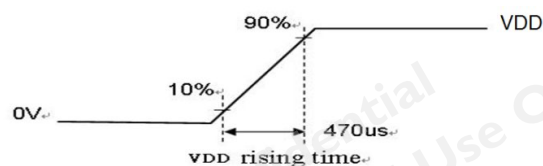
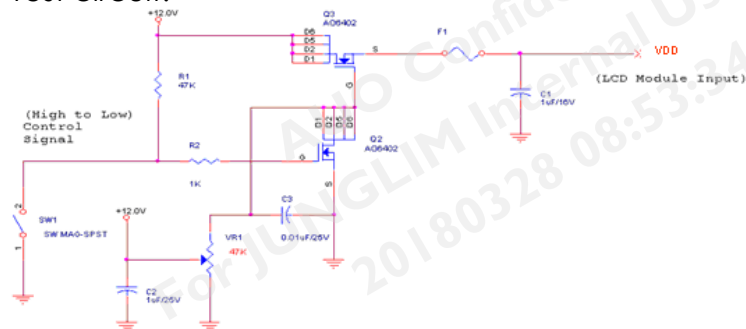
Symbol	Description	Min	Max	Unit	Remark
VDD	Power Supply Input Voltage	GND-0.3	14	[Volt]	Ta=25°C

3.3.2 Recommended Operating Condition

Symbol	Description	Min	Typ	Max	Unit	Remark
VDD	Power supply Input voltage	10.8	12	13.2	[Volt]	
IDD	Power supply Input Current (RMS)		0.45	0.54	[A]	VDD= 12V, Whie Pattern, Fv=60Hz
		-	0.52	0.62	[A]	VDD= 12V, Whie Pattern, Fv=75Hz
PDD	VDD Power Consumption		5.40	6.48	[Watt]	VDD= 12V, Whie Pattern, Fv=60Hz
		-	6.24	7.49	[Watt]	VDD= 12V, Whie Pattern, Fv=75Hz
IRush	Inrush Current	-	-	3	[A]	Note 3-1
VDDrp	Allowable VDD Ripple Voltage	-	-	500	[mV]	VDD= 12V, White Pattern, Fv=75Hz

Note 3-1: Inrush Current measurement:

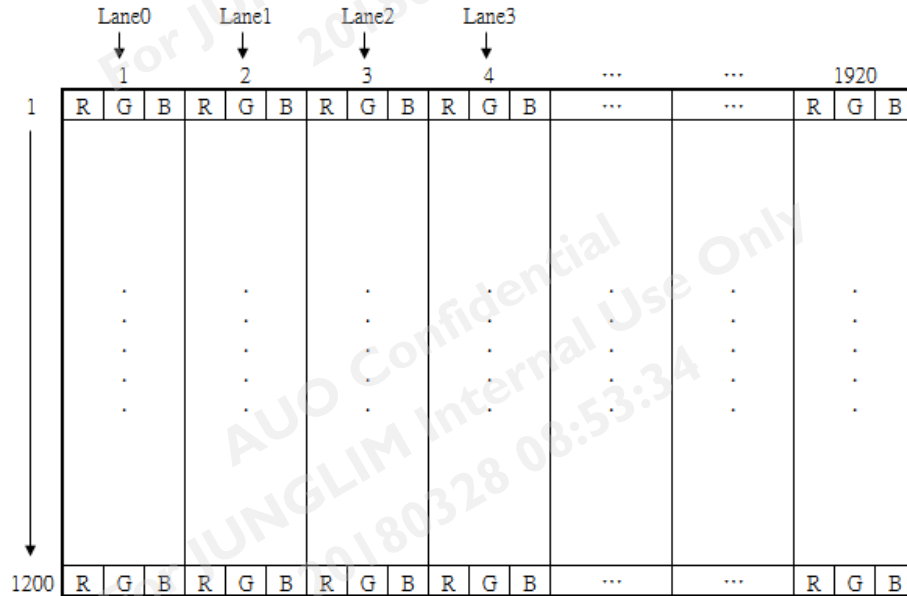
Test circuit:



The duration of VDD rising time: 470us.

3.4 Signal Characteristics

3.4.1 LCD Pixel Format



Note 3-2: The module use 4-Lanes eDP interface.

Lane0 : 1+4n pixel

Lane1 : 2+4n pixel

Lane2 : 3+4n pixel

Lane3 : 4+4n pixel

n=0~479

3.4.2 eDP Data Format

Lane0	Lane1	Lane2	Lane3
R1-9:2	R2-9:2	R3-9:2	R4-9:2
R1-1:0/G1-9:4	R2-1:0/G2-9:4	R3-1:0/G3-9:4	R4-1:0/G4-9:4
G1-3:0/B1-9:6	G2-3:0/B2-9:6	G3-3:0/B3-9:6	G4-3:0/B4-9:6
B1-5:0/R5-9:8	B2-5:0/R6-9:8	B3-5:0/R7-9:8	B4-5:0/R8-9:8
R5-7:0	R6-7:0	R7-7:0	R8-7:0
G5-9:2	G6-9:2	G7-9:2	G8-9:2
G5-1:0/B5-9:4	G6-1:0/B6-9:4	G7-1:0/B7-9:4	G8-1:0/B8-9:4
B5-3:0/R9-9:6	B6-3:0/R10-9:6	B7-3:0/R11-9:6	B8-3:0/R12-9:6
R9-5:0/G9-9:8	R10-5:0/G10-9:8	R11-5:0/G11-9:8	R12-5:0/G12-9:8
G9-7:0	G10-7:0	G11-7:0	G12-7:0
B9-9:2	B10-9:2	B11-9:2	B12-9:2
B9-1:0/R13-9:4	B10-1:0/R14-9:4	B11-1:0/R15-9:4	B12-1:0/R16-9:4
R13-3:0/G13-9:6	R14-3:0/G14-9:6	R15-3:0/G15-9:6	R16-3:0/G16-9:6
G13-5:0/B13-9:8	G14-5:0/B14-9:8	G15-5:0/B15-9:8	G16-5:0/B16-9:8
B13-7:0	B14-7:0	B15-7:0	B16-7:0
.	.	.	.
.	.	.	.
.	.	.	.
.	.	.	.

3.4.3 Color versus Input Data

The following table is for color versus input data (10bit). The higher the gray level, the brighter the color.

Color	Gary Level	Color Input Data																														Remark	
		RED data (MSB :R9, LSB :R0)										GREEN data (MSB :G9, LSB :G0)										BLUE data (MSB :B9, LSB :B0)											
		R9	R8	R7	R6	R5	R4	R3	R2	R1	R0	G9	G8	G7	G6	G5	G4	G3	G2	G1	G0	B9	B8	B7	B6	B5	B4	B3	B2	B1	B0		
Black	-	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
White	-	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	
L511	-	0	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	1	
Red	L0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Black
	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	
	L1023	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
Green	L0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Black
	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	
	L1023	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	
Blue	L0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Black	
	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	:	
	L1023	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	

3.4.4 eDP Specification (Follow as VESA DisplayPort Standard Version 1.1)

a. DisplayPort main link signal:

DisplayPort main link					
Symbol	Description	Min	Typ	Max	unit
Frequency	Main link Frequency	-	2.7	-	Gbps
UI	Unit Interval	-	370	-	ps
VCM	RX input DC Common Mode Voltage	-	0	-	[Volt]
VDiff _{P-P}	Peak-to-peak Voltage at a receiving Device	150	-	-	[mVolt]
Down_Spread_Freq	Link clock down spread frequency	30	-	33	KHz
Down_Spread_Amplitude	Link clock down spread amplitude	-	-	0.5	%

Point	Time (UI)	Voltage (V)
1	0.245	0
2	0.5	-75mV
3	0.755	0
4	0.5	-75mV

Figure 5.55.3 Downstream Device EYE Mask at Receiver Connector for HBR

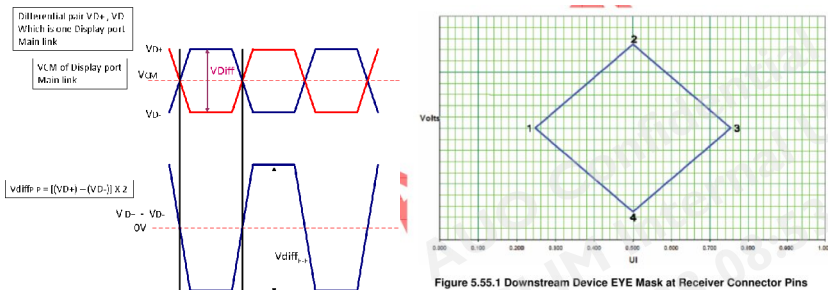
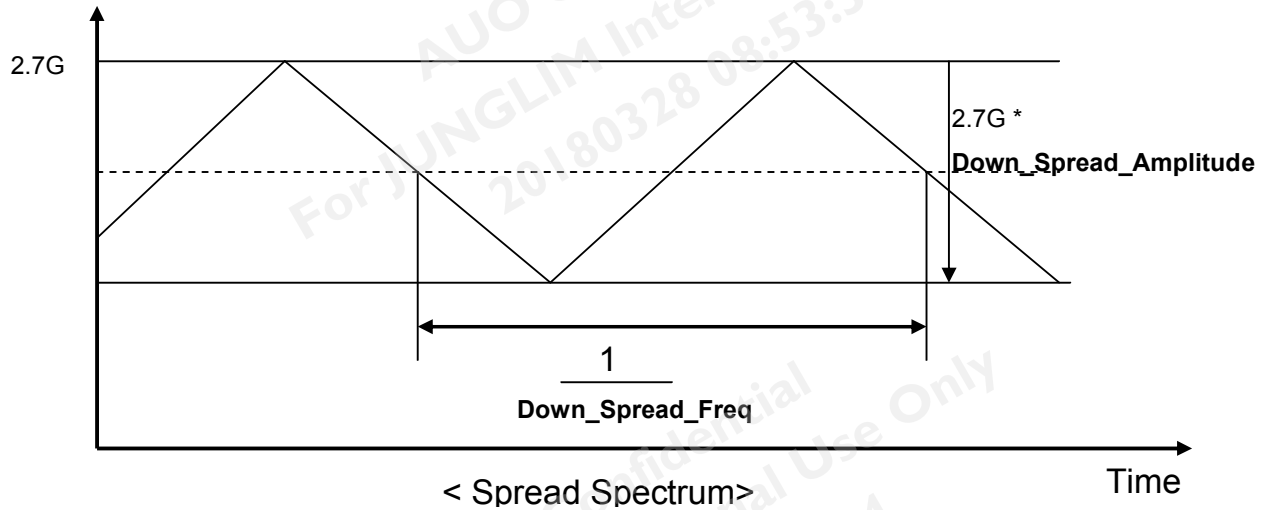


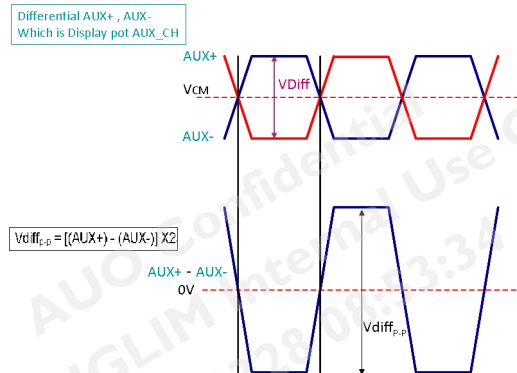
Figure 5.55.1 Downstream Device EYE Mask at Receiver Connector Pins

Freq(bps)



b. DisplayPort AUX_CH signal:

DisplayPort AUX_CH					
Symbol	Description	Min	Typ	Max	unit
VCM	AUX DC Common Mode Voltage	0	-	2.0	[Volt]
VDiff _{P-P}	AUX Peak-to-peak voltage at a receiving device	0.27	-	1.36	[Volt]



c. DisplayPort VHPD signal:

Display Port VHPD					
Symbol	Description	Min	Typ	Max	unit
VHPD	HPD Voltage	2.25	-	3.6	[Volt]

d. Intra-Pair skew

LRX-SKEW-INTRA_PAIR					
Symbol	Description	Min	Typ	Max	unit
LRX-SKE	Lane Intra-pair Skew Tolerance	-	-	50	[ps]



Product Specification

AU OPTRONICS CORPORATION

M240UAN02.1

W-INTRA_PAIR					
--------------	--	--	--	--	--

e. Inter-Pair Skew

LRX-SKEW-INTER_PAIR					
Symbol	Description	Min	Typ	Max	unit
LRX-SKEW-INTER_PAIR	Lane-to-Lane Skew at RX package pins	-	-	5200	[ps]

3.4.5 Input Timing Specification

It only support DE mode, and the input timing are shown as the following table.

Symbol	Description		Min.	Typ.	Max.	Unit	Remark
Tv	Vertical Section	Period	1220	1235	1985	Th	
Tdisp (v)		Active	1200	1200	1200	Th	
Tblk (v)		Blanking	20	35	785	Th	
Fv		Frequency	49	60	76	Hz	
Th	Horizontal Section	Period	2200	2280	3520	Tclk	
Tdisp (h)		Active	1920	1920	1920	Tclk	
Tblk (h)		Blanking	280	360	1600	Tclk	
Fh		Frequency	59.78	74.1	97.27	kHz	Note 3-3
Tclk	Pixel Clock	Period	4.67	5.92	7.60	ns	1/Fclk
Fclk		Frequency	131.52	168.95	214.00	MHz	Note 3-4
Link Rate per Lane			2.7			Gbps	

Note 3-3: The equation is listed as following. Please don't exceed the above recommended value.

$$Fh (\text{Min.}) = Fclk (\text{Min.}) / Th (\text{Min.})$$

$$Fh (\text{Typ.}) = Fclk (\text{Typ.}) / Th (\text{Typ.})$$

$$Fh (\text{Max.}) = Fclk (\text{Max.}) / Th (\text{Min.})$$

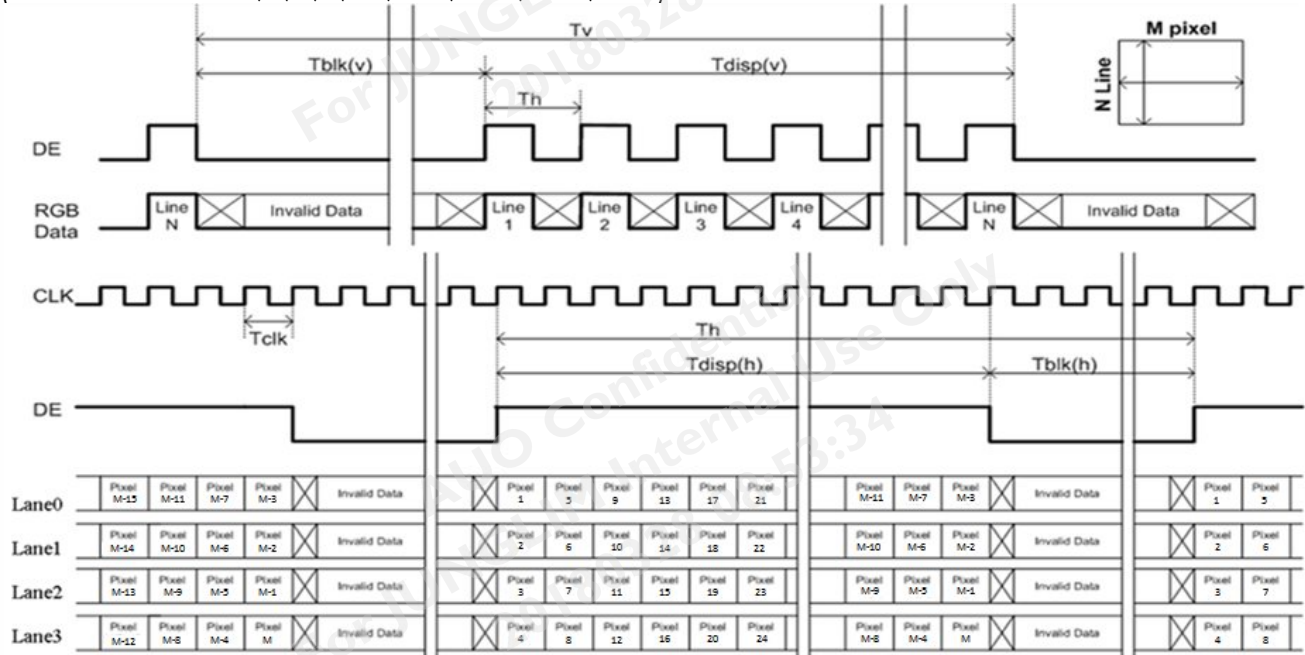
Note 3-4: The equation is listed as following. Please don't exceed the above recommended value.

$$Fclk (\text{Typ.}) = Fv (\text{Typ.}) \times Th (\text{Typ.}) \times Tv (\text{Typ.})$$

$$Fclk (\text{Min.}) \leq Fv \times Th \times Tv \leq Fclk (\text{Max.})$$

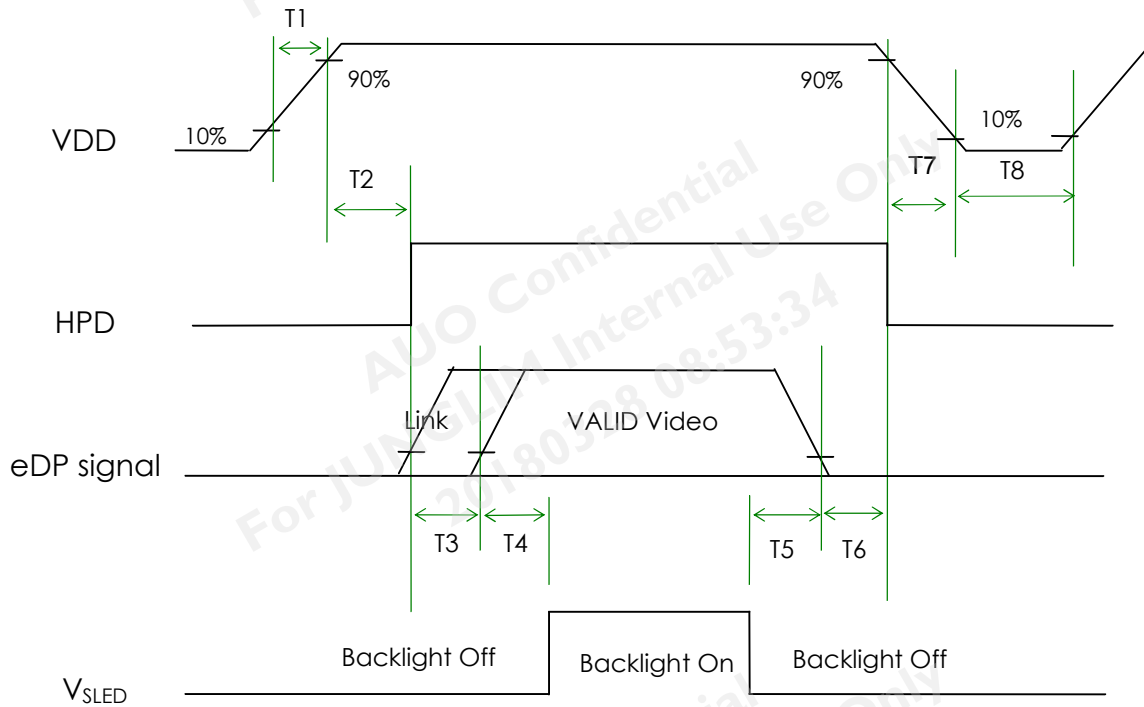
3.4.6 Input Timing Diagram

(Lane0~3 eDP data:1, 2, 3, 4,,1917, 1918, 1919, 1920)



3.5 Power ON/OFF Sequence

VDD power, eDP signal and backlight on/off sequence are as following. eDP signals from any system shall be Hi-Z state when VDD is off.



Power Sequence Timing

Symbol	Value			Unit	Remark
	Min.	Typ.	Max.		
T1	0.5	-	10	[ms]	
T2	0	-	200	[ms]	
T3	0	-	-	[ms]	Note 3-5
T4	500	-	-	[ms]	
T5	100	-	-	[ms]	
T6	0	-	50	[ms]	Note 3-6 Note 3-7
T7	0	-	200	[ms]	Note 3-7 Note 3-8
T8	1000	-	-	[ms]	

Note 3-5 : During T3 period , eDP link training time by customer's system.

Note 3-6 : Recommend setting T6 = 0ms to avoid electronic noise when VDD is off.



Product Specification

M240UAN02.1

AU OPTRONICS CORPORATION

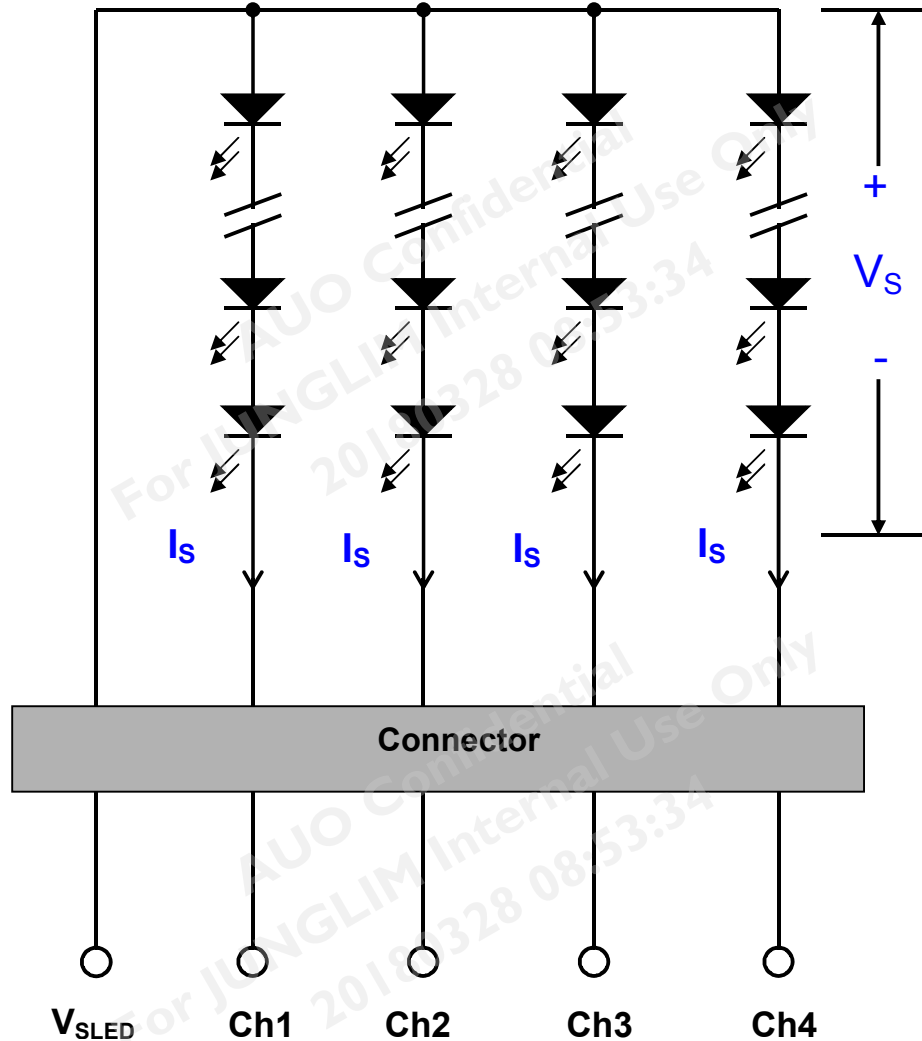
Note 3-7 : During T6 & T7 period , please keep the level of input eDP signals with Hi-Z state.

Note 3-8 : Voltage of VDD must decay smoothly after power-off.(customer system decide this value)

4 Backlight Unit

4.1 Block Diagram

The following shows the block diagram of the 24 inch Backlight Unit. And it includes 60 pcs LED in the LED light bar. (4 strings and 15 pcs LED of one string).



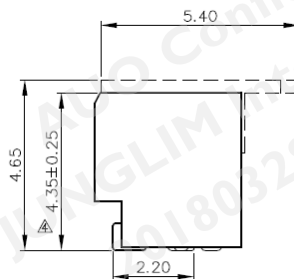
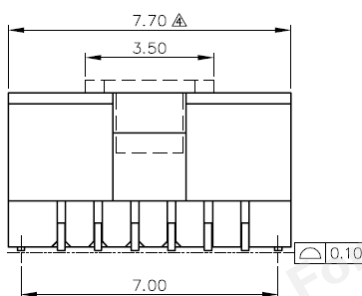
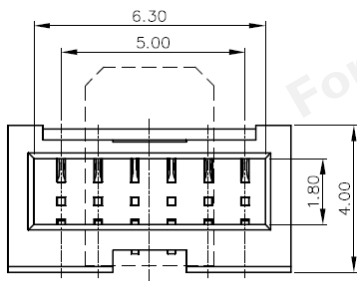
4.2 Interface Connection

4.2.1 Connector Type

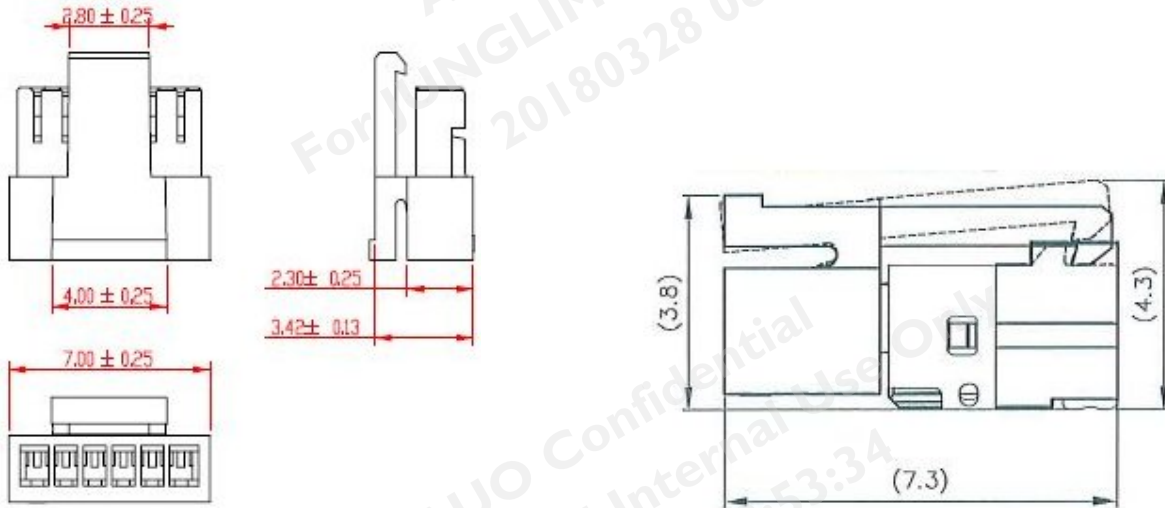
Backlight Connector	Manufacturer	ENTERY
	Part Number	3709K-Q06C-04L
Mating Connector	Manufacturer	ENTERY
	Part Number	H112K-P06N-00B (Non-Locking type) H112K-P06N-03B (Locking type)

Backlight Connector dimension:

$H \times V \times D = 7.7 \times 4.35 \times 4$, Pitch=1.0(unit=mm),use 6 pin connector

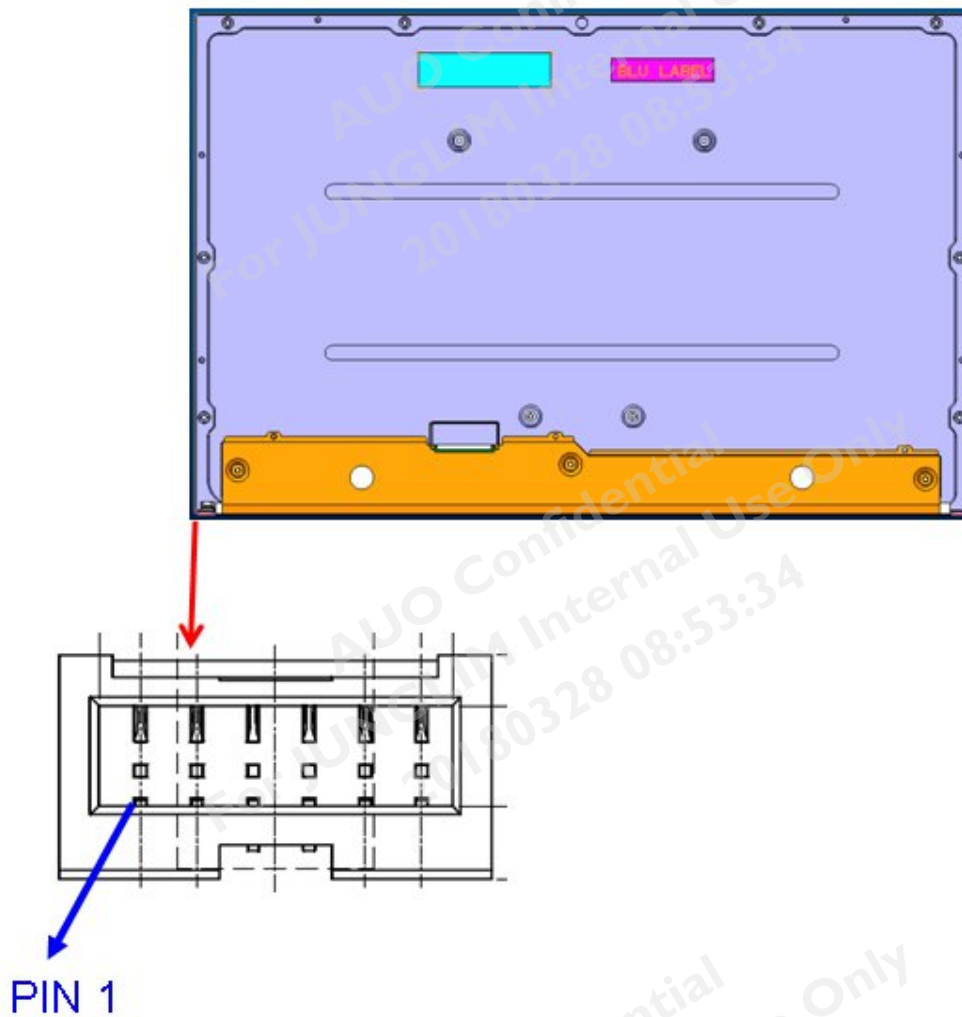


Mating Connector dimension:



4.2.2 Connector Pin Assignment

Pin#	Symbol	Description	Remark
1	Ch1	LED Current Feedback Terminal (Channel 1)	
2	Ch2	LED Current Feedback Terminal (Channel 2)	
3	V _{SLED}	LED Power Supply Voltage Input Terminal	
4	V _{SLED}	LED Power Supply Voltage Input Terminal	
5	Ch3	LED Current Feedback Terminal (Channel 3)	
6	Ch4	LED Current Feedback Terminal (Channel 4)	



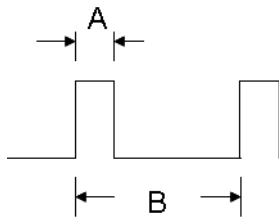
4.3 Electrical Characteristics

4.3.1 Absolute Maximum Rating

Permanent damage may occur if exceeding the following maximum rating.

(Ta=25°C)

Symbol	Description	Min	Max	Unit	Remark
Is	LED String Current	0	150	[mA]	100% duty ratio
			240	[mA]	Duty ratio ≤ 10% Pulse time=10 ms



Duty ratio= (A / B) X 100% ; (A: Pulse time, B: Period)

4.3.2 Recommended Operating Condition

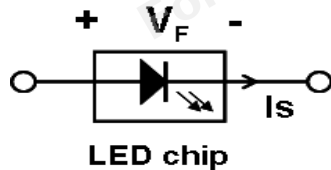
(Ta=25°C)

Symbol	Description	Min.	Typ.	Max.	Unit	Remark
Is	LED String Current	-	90	99	[mA]	100% duty ratio of LED chip Note 4-7
Vs	LED String Voltage	40.5	45	49.5	[Volt]	Is=90mA @ 100% duty ratio; Note 4-1, Note 4-5
ΔVs	Maximum Vs Voltage Deviation of light bar	-	-	3	[Volt]	Is=90mA @ 100% duty ratio; Note 4-2
P _{BLU}	LED Light Bar Power Consumption	-	16.2	17.8	[Watt]	Note 4-3
LT _{LED}	LED Life Time	30,000	-	-	[Hour]	Note 4-4
OVP	Over Voltage Protection in system board	110% Vsmax	-	-	[Volt]	Note 4-5

Note 4-1: $V_s (\text{Typ.}) = V_F (\text{Typ.}) \times \text{LED No. (one string)}$;

a. V_F : LED chip forward voltage, $V_F (\text{Min.})=2.7\text{V}$, $V_F (\text{Typ.})=3\text{V}$, $V_F (\text{Max.})=3.3\text{V}$

b. The same equation to calculate $V_s (\text{Min.})$ & $V_s (\text{Max.})$ for respective $V_F (\text{Min.})$ & $V_F (\text{Max.})$;



Note 4-2: $\Delta V_s (\text{Max.}) = \Delta V_F \times \text{LED No. (one string)}$;

a. ΔV_F : LED chip forward voltage deviation; (0.2 V , each Bin of LED V_F)

Note 4-3: $P_{BLU} (\text{Typ.}) = V_s (\text{Typ.}) \times I_s (\text{Typ.}) \times 4$; (4 is total String No. of LED Light bar)

$P_{BLU} (\text{Max.}) = V_s (\text{Max.}) \times I_s (\text{Typ.}) \times 4$;

Note 4-4: Definition of life time:

a. Brightness of LED becomes to 50% of its original value

b. Test condition: $I_s = 90\text{mA}$ and 25°C (Room Temperature)

Note 4-5: Recommendation for LED driver power design:

Due to there are electrical property deviation in LED & monitor set system component after long time operation. AUO strongly recommend the design value of LED driver board OVP (over voltage protection) should be 10% higher than max. value of LED string voltage (V_s) at least.

Note 4-6: AUO strongly recommend "Analog Dimming" method for backlight brightness control for Wavy Noise Free. Otherwise, recommend that Dimming Control Signal (PWM Signal) should be synchronized with Frame Frequency.

Note 4-7 Ensure that the LED light bar is not subjected either forward or reverse voltage while monitor set is on standby mode or not in use.

5 Reliability Test

AUO reliability test items are listed as following table. (Bare Panel only)

Items	Condition	Remark
Temperature Humidity Bias (THB)	Ta= 50°C , 80%RH, 300hours	
High Temperature Operation (HTO)	Ta= 50°C , 50%RH, 300hours	
Low Temperature Operation (LTO)	Ta= 0°C , 300hours	
High Temperature Storage (HTS)	Ta= 60°C , 300hours	
Low Temperature Storage (LTS)	Ta= -20°C , 300hours	
Vibration Test (Non-operation)	Acceleration: 1.5 Grms Wave: Random Frequency: 10 - 200 Hz Sweep: 30 Minutes each Axis (X, Y, Z)	
Shock Test (Non-operation)	Acceleration: 50 G Wave: Half-sine Active Time: 20 ms Direction: ±X, ±Y, ±Z (one time for each Axis)	
Thermal Shock Test (TST)	-20°C /30min, 60°C /30min, 100 cycles	Note 5-1
On/Off Test	On/10sec, Off/10sec, 30,000 cycles	
ESD (Electro Static Discharge)	Contact Discharge: ± 15KV, 150pF(330Ω) 1sec, 8 points, 25 times/ point.	Note 5-2
	Air Discharge: ± 15KV, 150pF(330Ω) 1sec 8 points, 25 times/ point.	
Altitude Test	Operation:18,000 ft Non-Operation:40,000 ft	

Note 5-1: a. A cycle of rapid temperature change consists of varying the temperature from -20°C to 60°C, and back again. Power is not applied during the test.

b. After finish temperature cycling, the unit is placed in normal room ambient for at least 4 hours before power on.

Note 5-2: EN61000-4-2, ESD class B: Certain performance degradation allowed

No data lost

Self-recoverable

No hardware failures.

ESD discharged points should avoid display area and periphery front bezel of display area.

Suggest point were 4 side parallel edge of display area surface.

Metal front bezel must cover half area of BM (black matrix) and metal front bezel



must connect with metal back bezel to protect source IC of panel by ESD damaged.

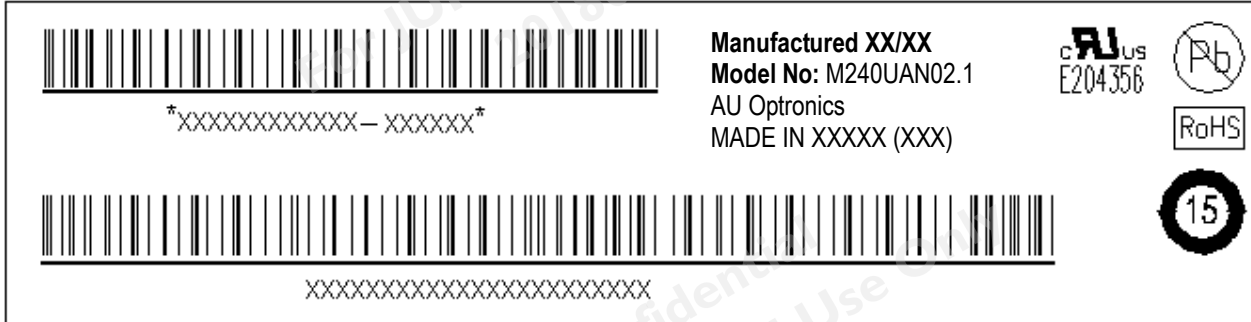
Note 5-3: Result Evaluation Criteria:

TFT-LCD panels test should take place after gradually cooling enough at room temperature.

In the normal application, there should be no particular problems that may affect the display function.

6 Shipping Label

The label is on the panel as shown below:



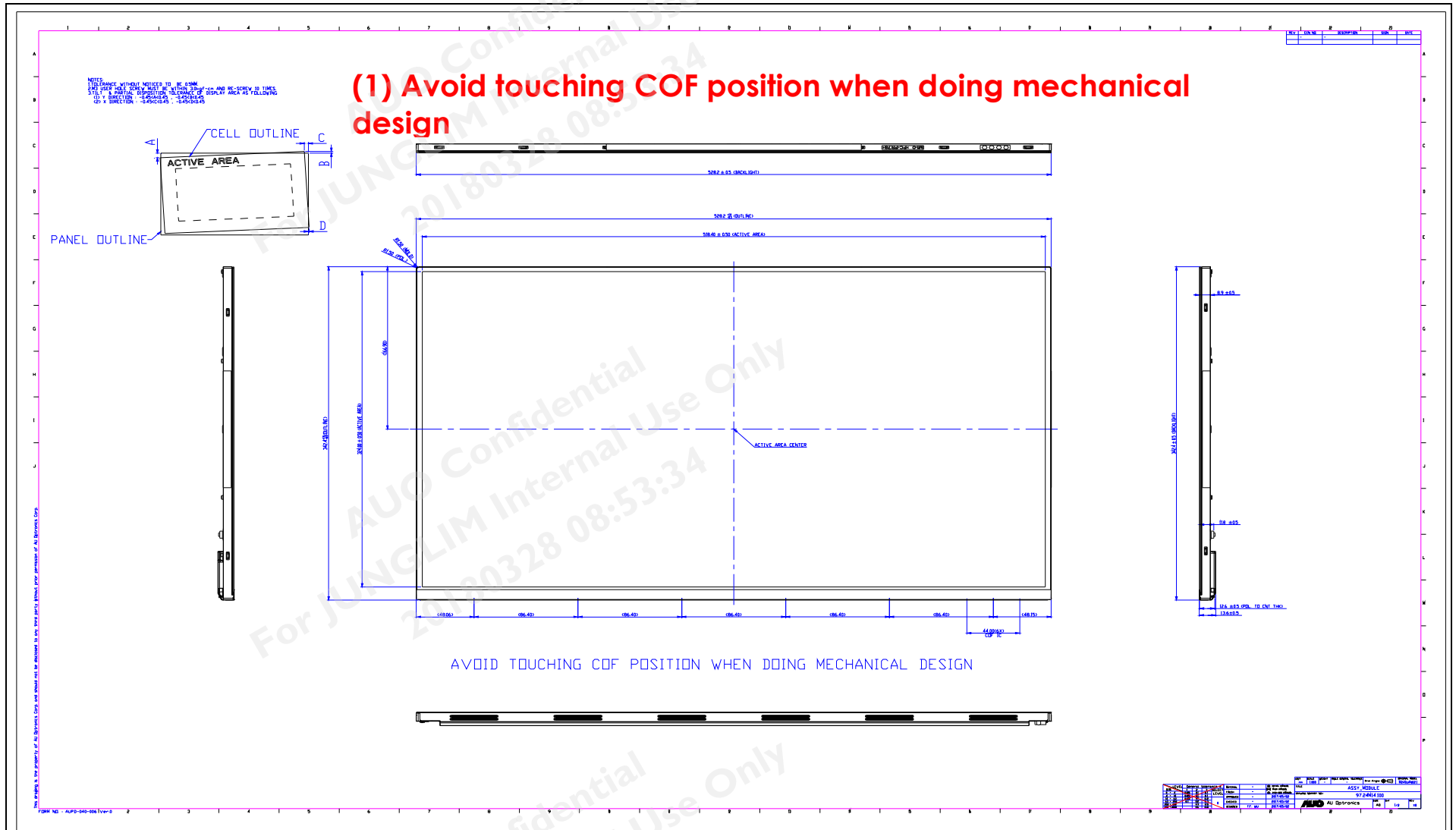
Note 6-1: For Pb Free products, AUO will add  for identification.

Note 6-2: For RoHS compatible products, AUO will add  for identification.

Note 6-3: For China RoHS compatible products, AUO will add  for identification.

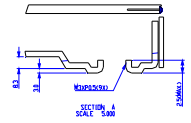
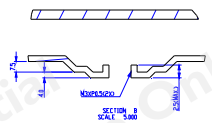
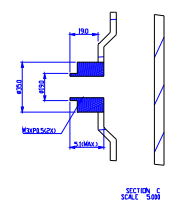
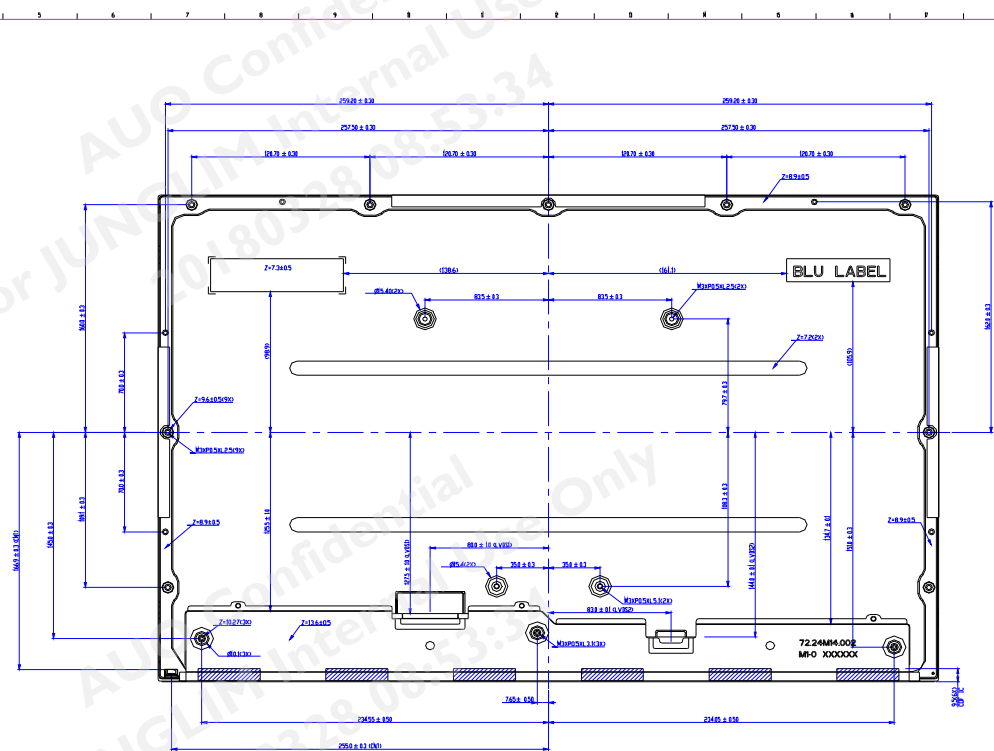
Note 6-4: The Green Mark will be presented only when the green documents have been ready by AUO Internal Green Team.

7 Mechanical Characteristics



Ver 1.1

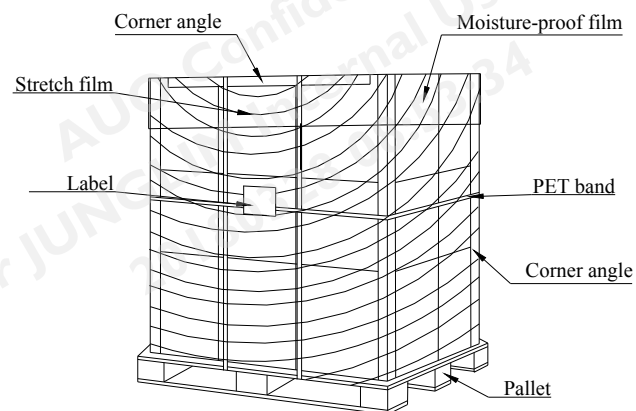
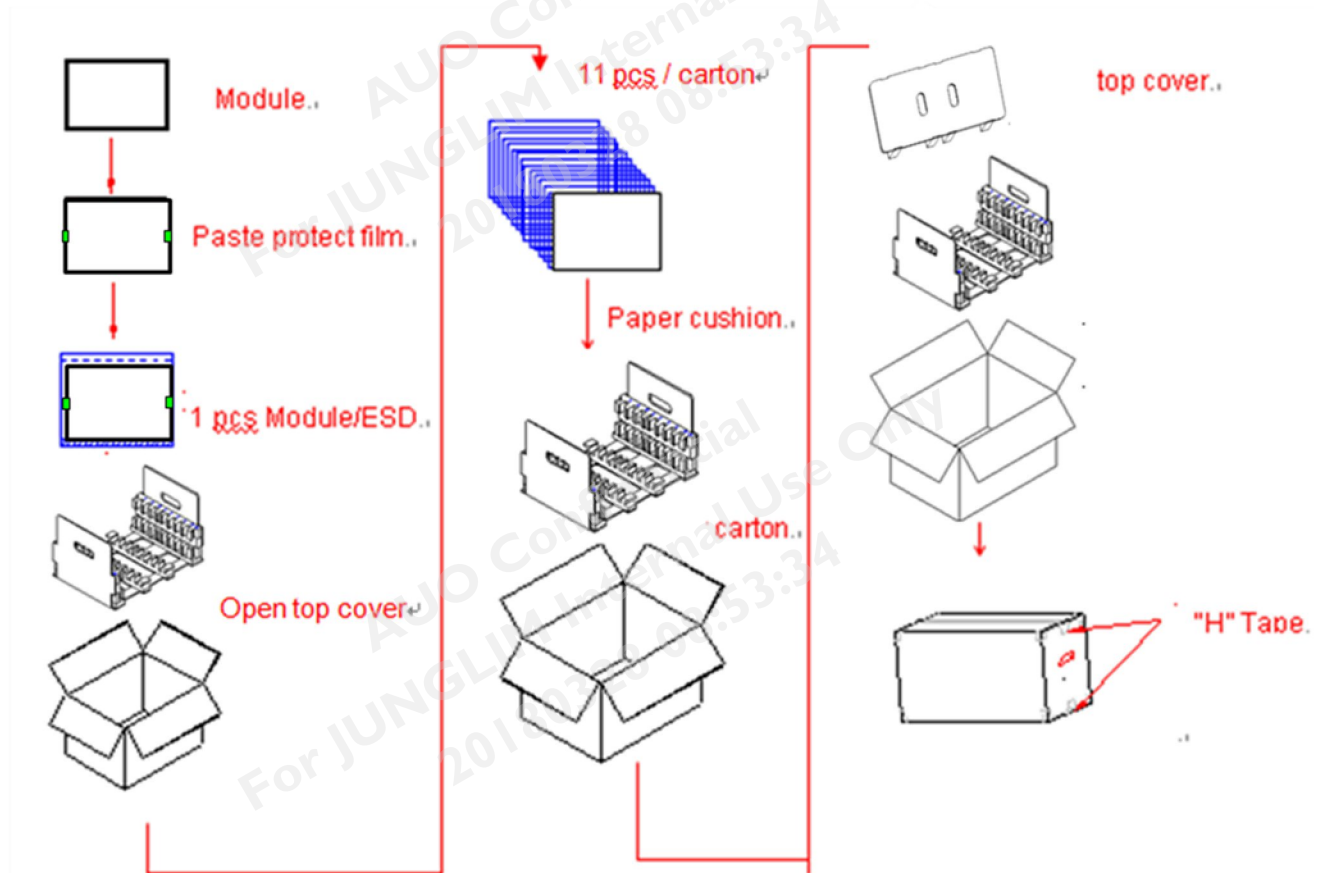
For JUNGLIM Internal Use Only
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REV	DATE	DESCRIPTION	BY	CHK
1	17/03	Initial Release	ASST. MICHAEL	
2	17/03	7224M4.002		
3	17/03	MP-O XXXXXXX		
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8 Packing Specification

8.1 Packing Flow



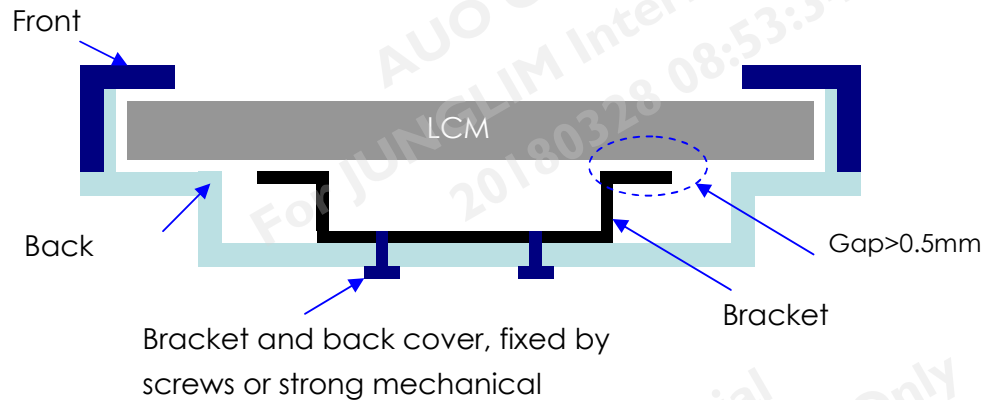
8.2 Pallet and shipment information

Item	Specification			Remark
	Q'ty	Dimension	Weight(kg)	
Panel	1	528.2(H) × 342.4(V) × 13.6(D) Typ..	2.42	
Cushion	1	-	0.6	
Box	1	617(L)mm x 371(W)mm x425(H)mm	1.5	without Panel & cushion
Packing Box	11 pcs/Box	617(L)mm x 371(W)mm x425(H)mm	29.93	with panel & cushion
Pallet	1	1315(L)mm x1150(W)mm x132(H)mm	17.00	
Pallet after Packing	18 boxes/pallet	1315(L)mm x1150(W)mm x1407(H)mm	533.96	

9 Design Guide for System

9.1 The gap between LCM and system rear bracket should be bigger than 0.5mm.

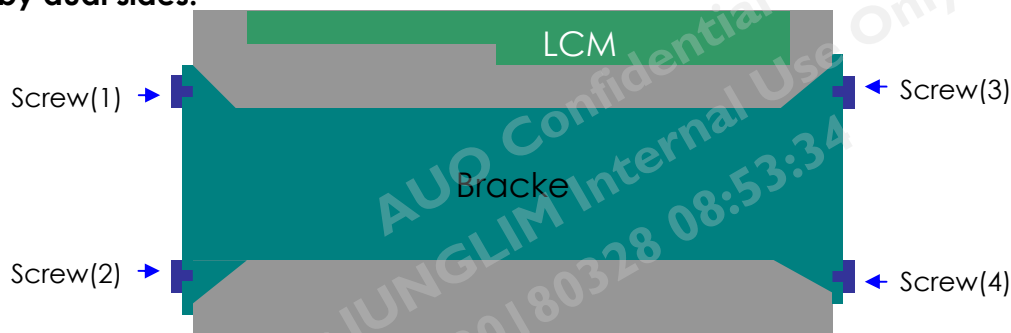
9.2 The system bracket should be fixed on back cover firmly.



9.3 The EMI gasket should be uniform and not push panel strongly.



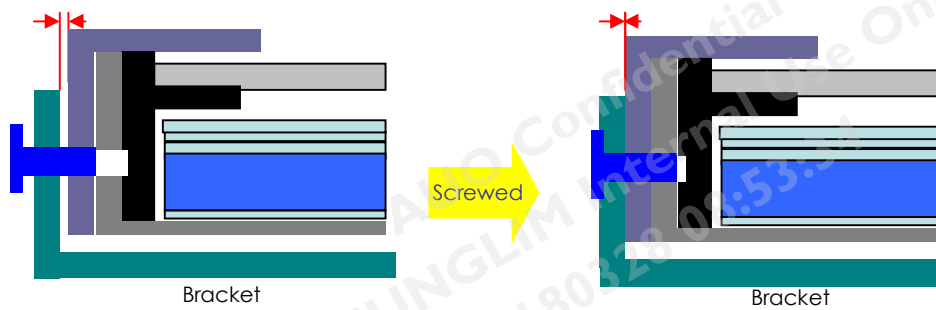
9.4 For stable assembly, the system bracket should use 4 screws to fix system and panel by dual sides.



9.5 The system bracket and panel should be in parallel with having no gap after inserting screws.

Proper and Parallel

0 gap and no mechanical damage



9.6 Avoid scratching LCM, the rib on system front-cover should not exceed the bottom edge of LCM's front-bezel.

